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(54) **LOW LATENCY ACTIVE NOISE CANCELLATION SYSTEM**

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This patent is subject to a terminal disclaimer.

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A61F 11/06 (2006.01)

(52) **U.S. Cl.**
USPC **381/71.1**; 381/71.6; 381/71.11

(58) **Field of Classification Search**
USPC 381/71.6, 71.7, 71.8, 71.11, 71.12,
381/71.1, 74, 309, 93, 94.1

See application file for complete search history.

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Primary Examiner — Vivian Chin

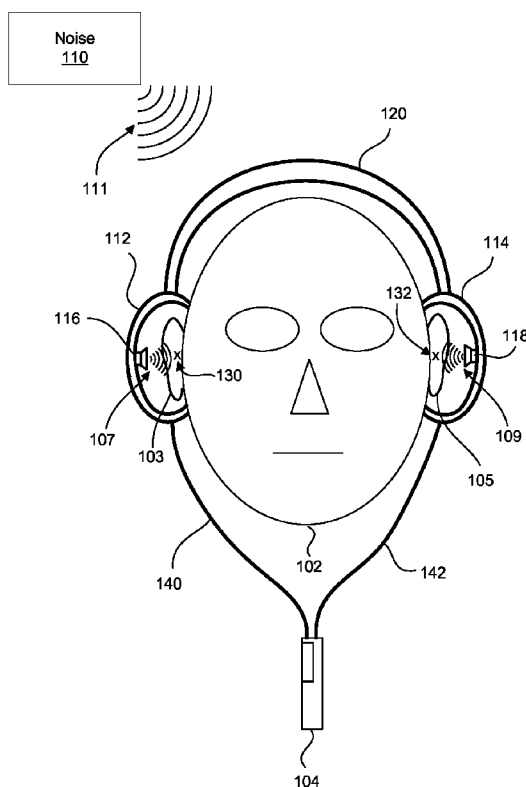
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(57) **ABSTRACT**

Systems and methods described herein provide for low latency active noise cancellation which alleviate the problems associated with analog filter circuitry. The present technology utilizes low latency digital signal processing techniques which overcome the high latency conventionally associated with conversion between the analog and digital domains. As a result, low latency active noise cancellation is performed utilizing digital filter circuitry which is not subject to the inaccuracies and drift of analog filter components. In doing so, the present technology provides robust, high quality active noise cancellation.

20 Claims, 8 Drawing Sheets



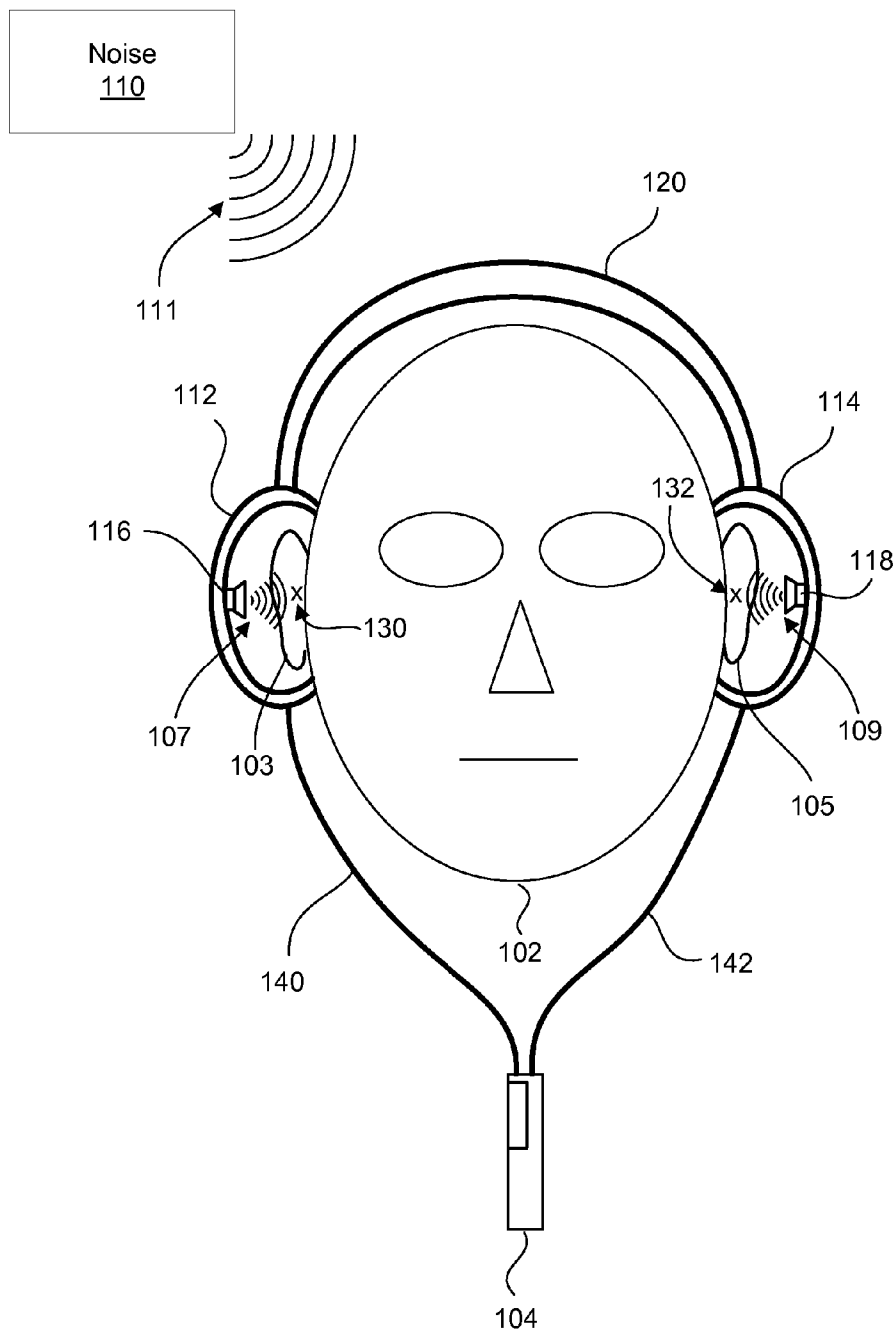


FIGURE 1

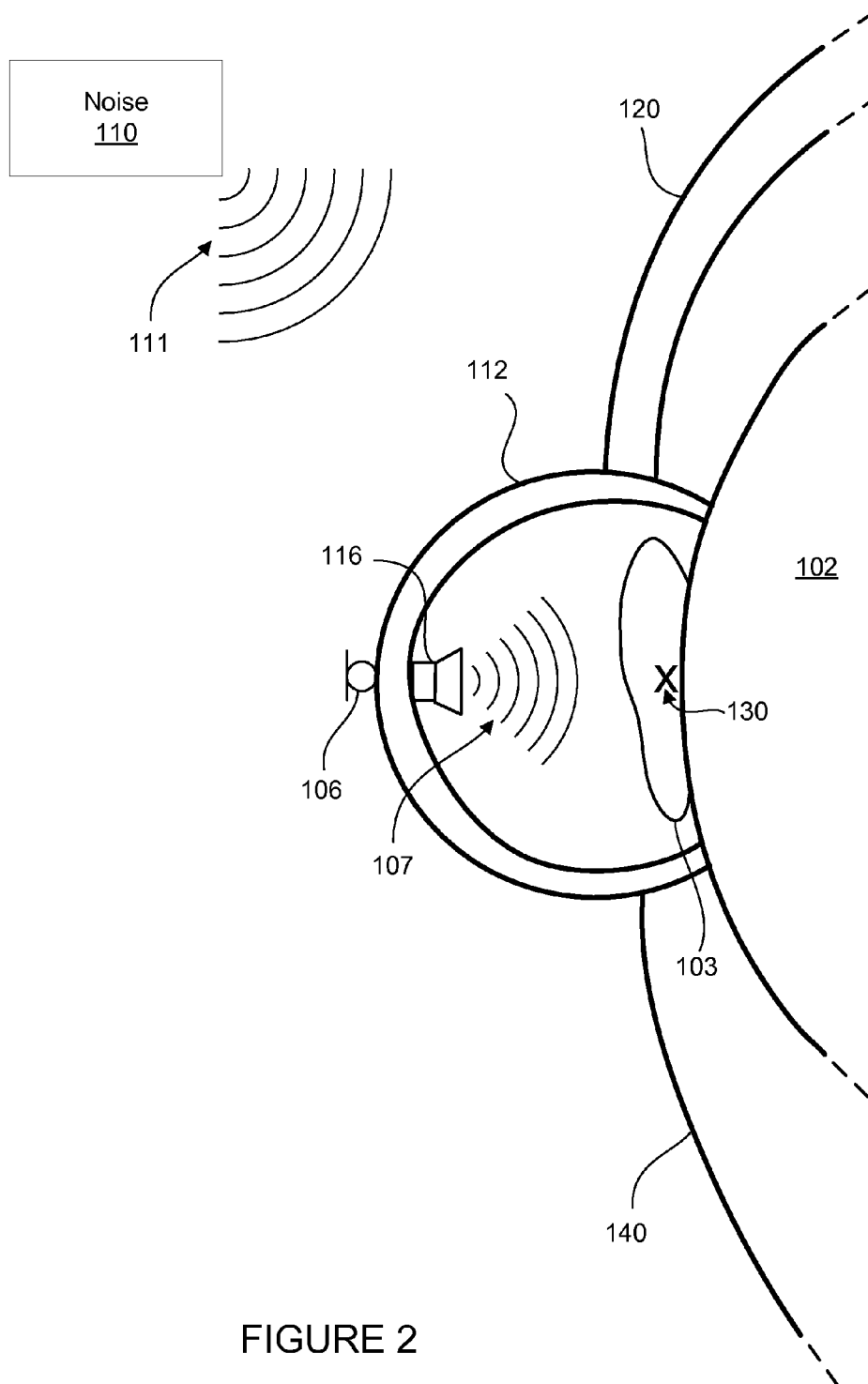


FIGURE 2

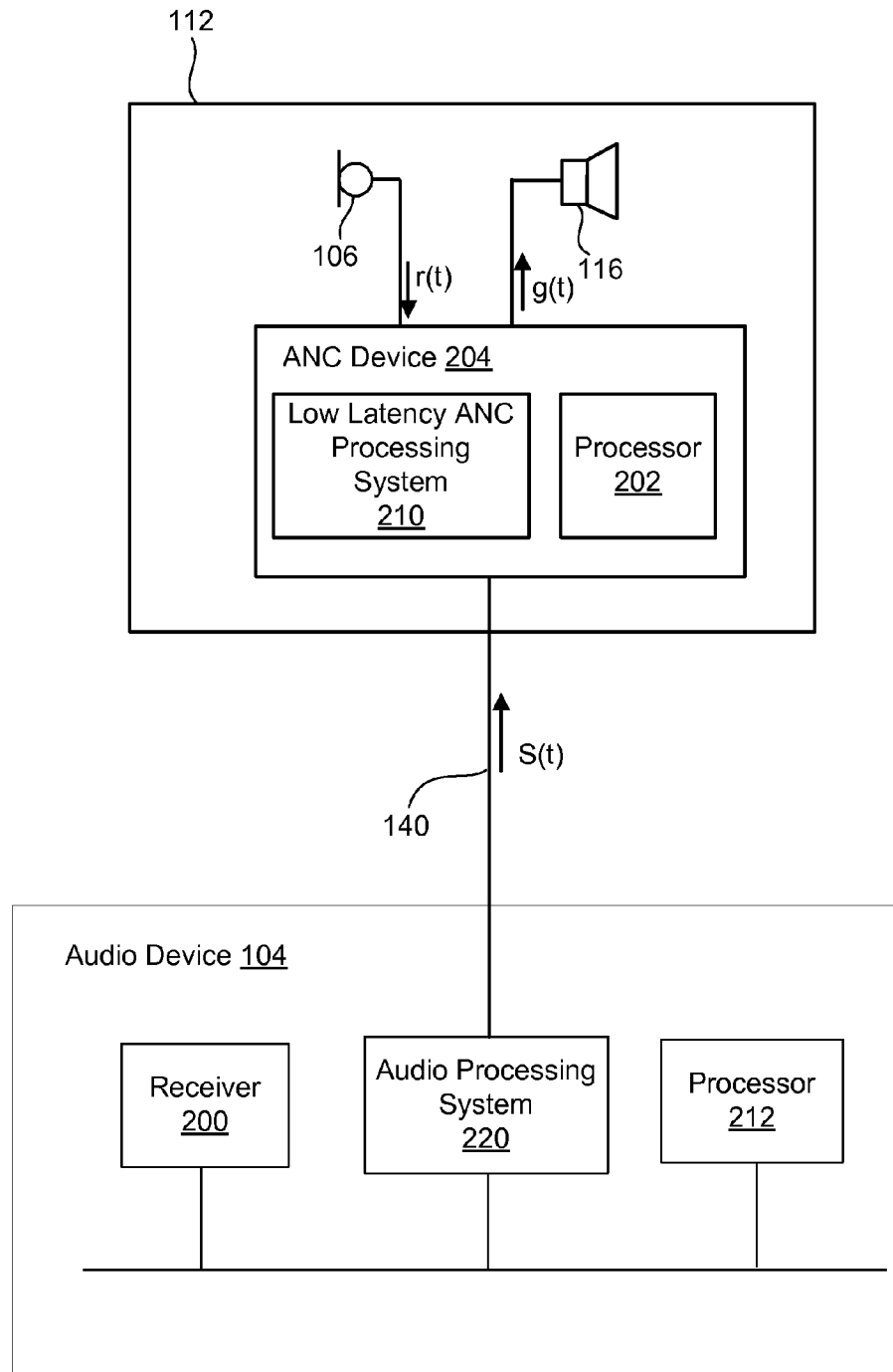


FIGURE 3

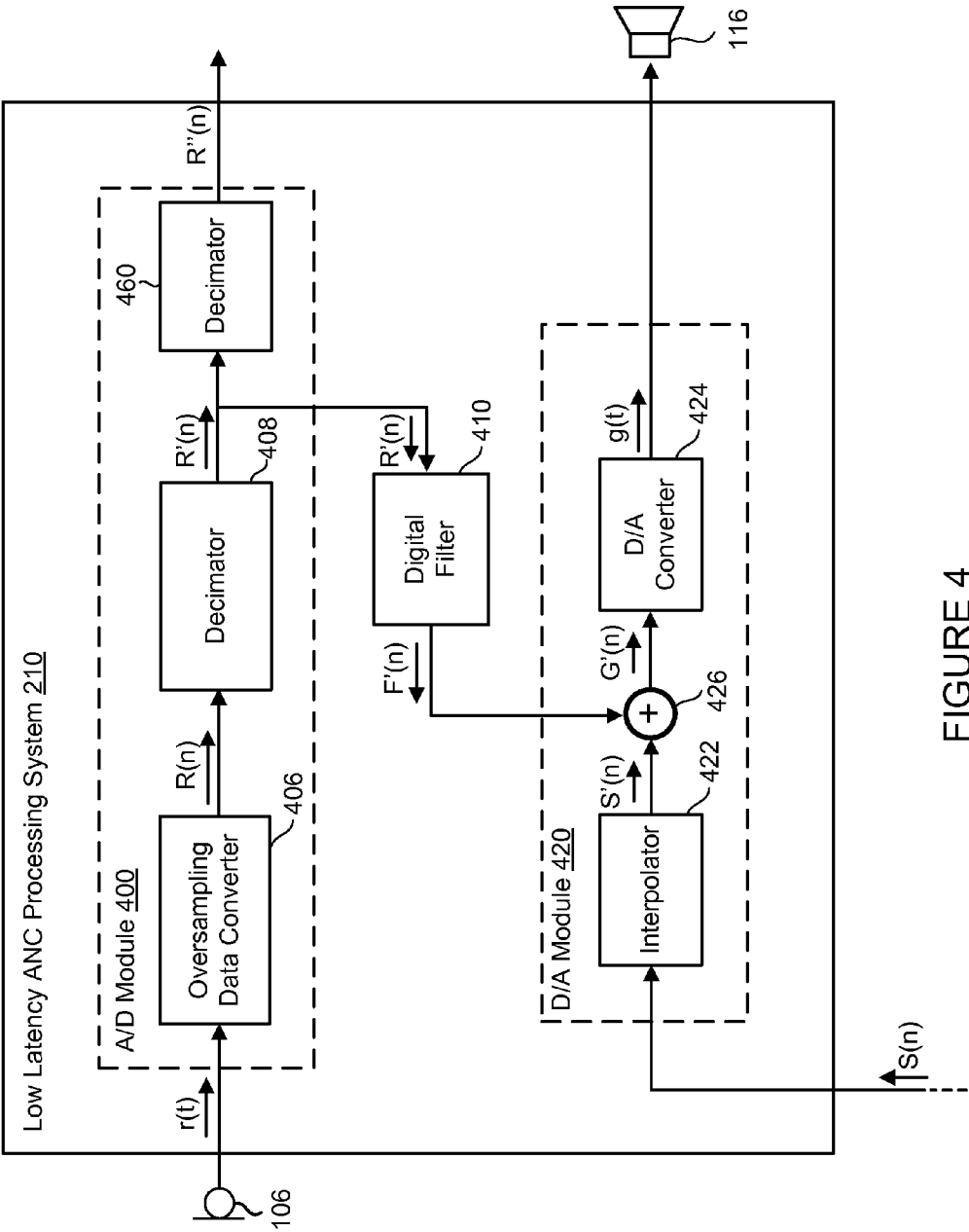


FIGURE 4

A/D Module 400

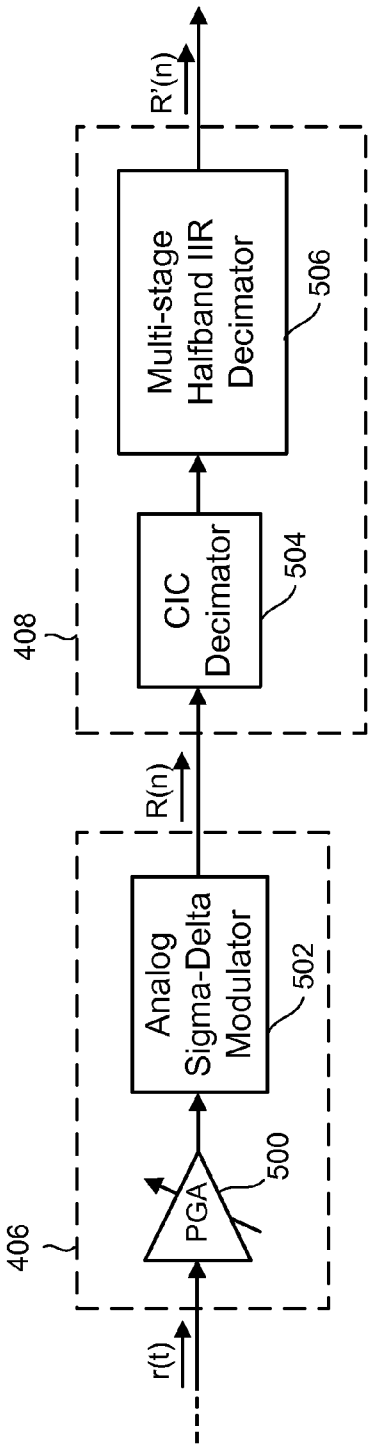


FIGURE 5

D/A Converter 424

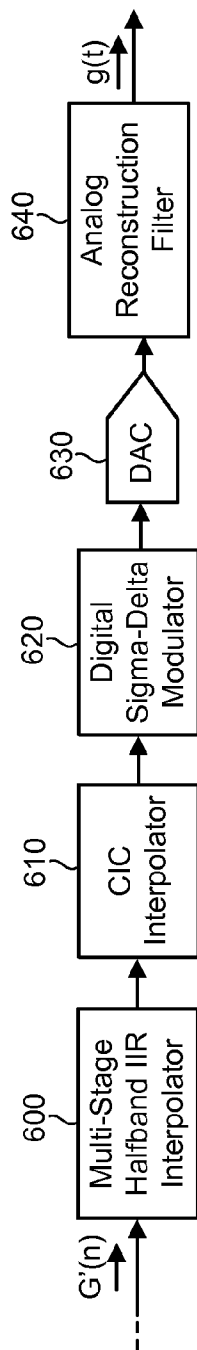


FIGURE 6

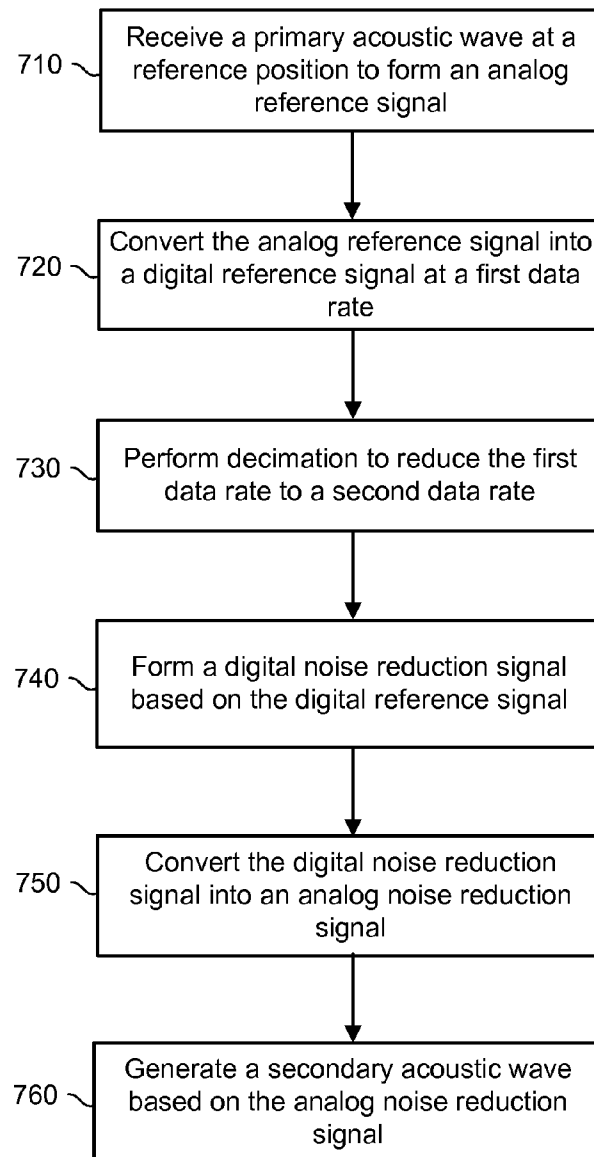
700

FIGURE 7

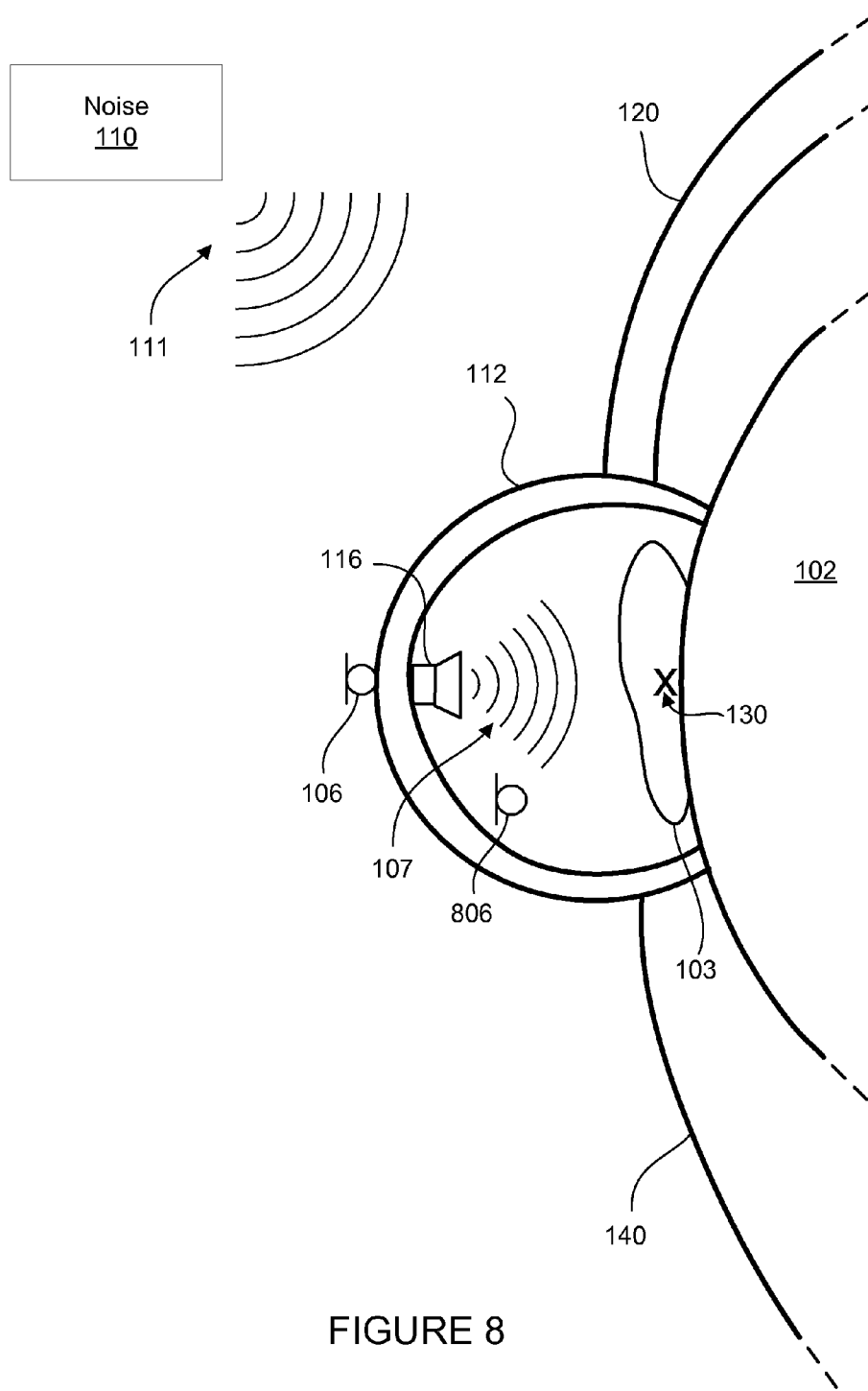


FIGURE 8

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LOW LATENCY ACTIVE NOISE CANCELLATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 61/286,117, filed on Dec. 14, 2009, entitled "Low Latency Active Noise Cancellation", which is incorporated by reference herein.

BACKGROUND

1. Field of the Invention

The present invention relates generally to audio processing, and more particularly to techniques for active noise cancellation.

2. Description of Related Art

An active noise cancellation (ANC) system in an earpiece based audio device can be used to reduce background noise. The ANC system forms a compensation signal adapted to cancel background noise at a listening position inside the earpiece. The compensation signal is provided to an audio transducer (e.g. a loudspeaker) which generates an "anti-noise" acoustic wave. The anti-noise acoustic wave is intended to attenuate or eliminate the background noise at the listening position via destructive interference, so that only the desired audio remains. Consequently, the combination of the anti-noise acoustic wave and the background noise at the listening position results in cancellation of both and hence a reduction in noise.

ANC systems may generally be divided into feedforward ANC systems and feedback ANC systems. In a typical feedforward ANC system, a reference microphone provides a reference signal based on the background noise captured at a reference position. The reference signal is then used by the ANC system to predict the background noise at the listening position so that it can be cancelled. Typically, this prediction utilizes a transfer function which models the acoustic path from the reference position to the listening position. Active noise cancellation is then performed to form a compensation signal adapted to cancel the noise, whereby the reference signal is filtered based on the transfer function.

The performance of the ANC system is constrained by the latency (or delay) introduced during the formation of the compensation signal. The latency limits the amount of noise attenuation achievable by the ANC system. For feedforward systems, excessive latency makes the anti-noise signal arrive too late to effectively cancel the noise signal, resulting in unsatisfactory cancellation at higher frequencies. For feedback systems, excessive latency can cause the closed-loop system to become unstable when the feedback gain is increased, effectively limiting the gain to a small value, which results in degraded noise attenuation performance. In either case, the resulting residual noise can interfere with the listening experience of desired sound, and is annoying. In some instances, the latency may result in the generation of an anti-noise acoustic wave that constructively interferes with the background noise at the listening position. In such a case, the combination of the anti-noise acoustic wave and the background noise may result in an increase in the noise at the listening position, rather than a decrease.

In order to achieve a relatively low latency, an ANC system may be implemented using analog filter circuitry. The analog circuitry filters and inverts the analog reference signal received from the reference microphone to form an analog compensation signal which is then provided to the loud-

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speaker. Although low latency can be achieved, the use of analog filter circuitry to perform active noise cancellation results in a number of drawbacks. For example, it can be difficult to achieve high precision or accuracy using analog filter components due to component variation. As a result, the component variation limits the overall noise cancellation performance of the ANC system. In addition, analog filter components are susceptible to drift and aging which can cause the performance to worsen over time. Finally, it can be difficult to change component values to adapt to various situations or to provide the user more flexibility in the amount or the nature of the noise attenuation, which makes analog circuitry less flexible in practice than digital solutions.

It is therefore desirable to provide low latency active noise cancellation techniques which can also address the problems associated with analog filter circuitry.

SUMMARY

Systems and methods described herein provide for low latency active noise cancellation which alleviate the problems associated with analog filter circuitry. The present technology utilizes low latency digital signal processing techniques which overcome the high latency conventionally associated with conversion between the analog and digital domains. As a result, low latency active noise cancellation is performed utilizing digital filter circuitry which is not subject to the inaccuracies and drift of analog filter components. In doing so, the present technology provides robust, flexible, and high quality active noise cancellation.

A method for reducing an acoustic energy level at a listening position as described herein includes receiving a primary acoustic wave at a reference position to form an analog reference signal. The analog reference signal is converted into a digital reference signal using an oversampling data converter. A digital noise reduction signal is then formed based on the digital reference signal. The digital noise reduction signal is then converted into an analog noise reduction signal. A secondary acoustic wave is then generated based on the analog noise reduction signal. The secondary acoustic wave is adapted to reduce the acoustic energy level at the listening position.

A system for reducing an acoustic energy level at a listening position as described herein includes a reference microphone to receive a primary acoustic wave at a listening position. The system also includes a noise cancellation module to convert the analog reference signal into a digital reference signal using an oversampling data converter. The noise cancellation module then forms a digital noise reduction signal based on the digital reference signal, and converts the digital noise reduction signal into an analog noise reduction signal. The system further includes an audio transducer to generate a secondary acoustic wave based on the analog noise reduction signal, the second acoustic wave adapted to reduce the acoustic energy level at the listening position.

A computer readable storage medium as described herein has embodied thereon a program executable by a processor to perform a method for reducing an acoustic energy level at a listening position as described above.

Other aspects and advantages of the present invention can be seen on review of the drawings, the detailed description, and the claims which follow.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an illustration of an environment in which embodiments of the present technology may be used.

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FIG. 2 is an expanded view of an exemplary earpiece of a headset.

FIG. 3 is a block diagram of an exemplary audio device coupled to an exemplary earpiece of the headset.

FIG. 4 is a block diagram of an exemplary low latency active noise cancellation (ANC) processing system for performing active noise cancellation as described herein.

FIG. 5 is a block diagram of an exemplary analog-to-digital converter module.

FIG. 6 is a block diagram of an exemplary digital-to-analog converter module.

FIG. 7 is a flow chart of an exemplary method for performing active noise cancellation.

FIG. 8 is an expanded view of a second exemplary earpiece of a headset.

DETAILED DESCRIPTION

Systems and methods described herein provide for low latency active noise cancellation which alleviate the problems associated with analog filter circuitry. The present technology utilizes low latency digital signal processing techniques which overcome the high latency conventionally associated with conversion between the analog and digital domains. As a result, low latency active noise cancellation is performed utilizing digital filter circuitry which is not subject to the inaccuracies and drift of analog filter components. In doing so, the present technology provides robust, flexible and high quality active noise cancellation.

Embodiments of the present technology may be practiced on any earpiece-based audio device that is configured to receive and/or provide audio such as, but not limited to, cellular phones, MP3 players, phone handsets and headsets. While some embodiments of the present technology will be described in reference to operation on a cellular phone, the present technology may be practiced on any audio device.

FIG. 1 is an illustration of an environment in which embodiments of the present technology may be used. An audio device 104 may act as a source of audio content to a headset 120 which can be worn over or in the ears 103, 105 of a user 102. The audio content provided by the audio device 104 may for example be stored on a storage media such as a memory device, an integrated circuit, a CD, a DVD, etc for playback to the user 102. The audio content provided by the audio device 104 may include a far-end acoustic signal received over a communications network, such as speech of a remote person talking into a second audio device. The audio device 104 may provide the audio content as mono or stereo acoustic signals to the headset 120 via one or more audio outputs. As used herein, the term "acoustic signal" refers to a signal derived from or based on an acoustic wave corresponding to actual sounds, including acoustically derived electrical signals which represent an acoustic wave.

In the illustrated embodiment, the exemplary headset 120 includes a first earpiece 112 positionable on or in the ear 103 of the user 102, and a second earpiece 114 positionable on or in the ear 105 of the user 102. Alternatively, the headset 120 may include a single earpiece. The term "earpiece" as used herein refers to any sound delivery device positionable on or in a person's ear, such as for example an ear bud, head phone, or other speaker mechanism.

The audio device 104 may be coupled to the headset 120 via one or more wires, a wireless link, or any other mechanism for communication of information. In the illustrated embodiment, the audio device 104 is coupled to the first earpiece 112 via wire 140, and is coupled to the second earpiece 114 via wire 142.

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The first earpiece 112 includes an audio transducer 116 which generates an acoustic wave 107 proximate the ear 103 of the user 102 in response to a first acoustic signal. The second earpiece 114 includes an audio transducer 118 which generates an acoustic wave 109 proximate the ear 105 of the user 102 in response to a second acoustic signal. Each of the audio transducers 116 and 118 may for example be a loudspeaker, or any other type of audio transducer which generates an acoustic wave in response to an electrical signal.

As described below, the first acoustic signal includes a desired signal such as the audio content provided by the audio device 104. The first acoustic signal also includes a first noise reduction signal adapted to cancel undesired background noise at a first listening position 130 using the techniques described herein. Similarly, the second acoustic signal includes a desired signal such as the audio content provided by the audio device 104. The second acoustic signal also includes a second noise reduction signal adapted to cancel undesired background noise at a second listening position 132 using the techniques described herein. In some alternative embodiments, the desired signals may be omitted.

As shown in FIG. 1, an acoustic wave (or waves) 111 will also be generated by noise 110 in the environment surrounding the user 102. Although the noise 110 is shown coming from a single location in FIG. 1, the noise 110 may include any sounds coming from one or more locations that differ from the location of the audio transducers 116, 118 and may include reverberations and echoes. The noise 110 may be stationary, non-stationary, and/or a combination of both stationary and non-stationary noise.

The total acoustic wave at the first listening position 130 is a superposition of the acoustic wave 107 generated by the audio transducer 116 and the acoustic wave 111 generated by the noise 110. The first listening position 130 may for example be in front of the eardrum of ear 103, where the user 102 will hear it. As described herein, the portion of the acoustic wave 107 due to the first noise reduction signal is configured to destructively interfere with the acoustic wave 111 at the first listening position 130. In other words, the combination of the portion of the acoustic wave 107 due to the first noise reduction signal and the acoustic wave 111 due to the noise 110 at the first listening position 130 results in cancellation of both and hence a reduction in the acoustic energy level of noise at the first listening position 130. As a result, the portion of the acoustic wave 107 due to the desired audio signal remains at the first listening position 130, where the user 102 will hear it.

Similarly, the total acoustic wave at the second listening position 132 is a superposition of the acoustic wave 109 generated by the audio transducer 118 and the acoustic wave 111 generated by the noise 110. The second listening position 132 may for example be in front of the eardrum of the ear 105. Using the techniques described herein, the portion of the acoustic wave 109 due to the second noise reduction signal is configured to destructively interfere with the acoustic wave 111 at the second listening position 132. In other words, the combination of the portion of the acoustic wave 109 due to the second noise reduction signal and the acoustic wave 111 due to the noise 110 at the second listening position 132 results in cancellation of both. As a result, the portion of the acoustic wave 109 due to the desired signal remains at the second listening position 132, where the user 102 will hear it.

The first earpiece 112 is representative of the first and second earpieces 112, 114. FIG. 2 is an expanded view of the first earpiece 112. In the following discussion, active noise cancellation techniques are described herein with reference to the first earpiece 112. It will be understood that the techniques

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described herein can also be extended to the second earpiece **114** to perform active noise cancellation at the second listening position **132**.

As shown in FIG. 2, the first earpiece **112** includes a reference microphone **106** at a reference position on the outside of the first earpiece **112**. Alternatively, the reference microphone **106** may be positioned within the first earpiece **112**.

The acoustic wave **111** due to the noise **110** is received by the reference microphone **106** and converted into an analog reference signal $r(t)$. As used herein, an “analog signal” is a signal whose value at any given moment in time can take on any value within a continuous range of values. As used herein, a “digital signal” is a signal whose value at any given moment in time can take on only a finite number of discrete values within a range of values and which is defined over a discrete set of time samples.

As described below, the analog reference signal $r(t)$ is converted into a decimated digital reference signal $R'(n)$ using an oversampling data converter such as a sigma-delta modulator. The digital reference signal $R'(n)$ is then filtered using a digital filter to form a digital noise reduction signal $F'(n)$. The digital filter is based on a transfer function which models the acoustic path from the location of the reference microphone **106** to the first listening position **130**. The transfer function may incorporate characteristics of the acoustic path, such as one or more of an amplitude, phase shift and time delay from the reference microphone **106** to the first listening position **130**. The transfer function can also model the reference microphone **106** response, the audio transducer **116** response, and the acoustic path from the audio transducer **116** to the first listening position **130**.

An analog electric signal $g(t)$, which is formed by converting the digital noise reduction signal $F'(n)$ and optionally a digital desired signal $S(n)$ from the audio device **104**, is then provided to the audio transducer **116**. In other words, the analog electric signal $g(t)$ is a superposition of an analog noise reduction signal $f'(t)$ corresponding to the digital noise reduction signal $F'(n)$, and an analog desired signal $s(t)$ corresponding to the digital desired signal $S(n)$. Active noise cancellation is then performed at the first listening position **130**, whereby the audio transducer **116** generates the acoustic wave **107** in response to the analog electric signal $g(t)$.

FIG. 3 is a block diagram of an exemplary audio device **104** coupled to an exemplary first earpiece **112** of the headset **120**. In the illustrated embodiment the audio device **104** is coupled to the first earpiece **112** via wire **140**. The audio device **104** may be coupled to the second earpiece **114** in a similar manner. Alternatively, other mechanisms may be used to couple the audio device **104** to the headset **120**.

In the illustrated embodiment, the audio device **104** includes a receiver **200**, a processor **212** and an audio processing system **220**. The audio device **104** may further include additional or other components necessary for operation of the audio device **104**. Similarly, the audio device **104** may include fewer components that perform similar or equivalent functions to those depicted in FIG. 3. In some embodiments, the audio device **104** includes one or more microphones and/or one or more output devices.

Processor **212** may execute instructions and modules stored in a memory (not illustrated in FIG. 3) in the audio device **104** to perform various operations. Processor **212** may include hardware and software implemented as a processing unit, which may process floating operations and other operations for the processor **212**.

The exemplary receiver **200** is configured to receive a signal from a communications network. In some embodiments, the receiver **200** may comprise an antenna device. The

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signal may then be forwarded to the audio processing system **220**, and provided as audio content to the user **102** via the headset **120** in conjunction with active noise cancellation as described herein.

The audio processing system **220** is configured to provide desired audio content to the first earpiece **112** in the form of digital desired audio signal $S(n)$. Similarly, the audio processing system **220** is configured to provide desired audio content to the second earpiece **114** in the form of a second digital desired audio signal (not illustrated). The audio content may be retrieved for example from data stored on a storage media such as a memory device, an integrated circuit, a CD, a DVD, etc., for playback to the user **102**. The audio content may include a far-end acoustic signal received over a communications network, such as speech of a remote person talking into a second audio device. The desired audio signals may be provided as mono or stereo signals.

An example of the audio processing system **220** in some embodiments is disclosed in U.S. patent application Ser. No. 12/832,920 filed on Jul. 8, 2010 and entitled “Multi-Microphone Robust Noise Suppression”, which is incorporated herein by reference.

The exemplary earpiece **112** includes the reference microphone **106**, audio transducer **116**, and active noise cancellation (ANC) device **204**. In some embodiments, more than two monitoring microphones may be used.

The ANC device **204** includes processor **202** and ANC processing system **210**. The processor **202** may execute instructions and modules stored in a memory (not illustrated in FIG. 3) in the ANC device **204** to perform various operations, including low latency active noise cancellation as described herein.

The ANC processing system **210** is configured to receive the reference signal $r(t)$ from the reference microphone **106**, and process the signal. Processing includes performing active noise cancellation as described herein. The ANC processing system **210** is discussed in more detail below.

In the illustrated embodiment, the active noise cancellation techniques are carried out by the low latency ANC processing system **210** of the ANC device **204**. Thus, in the illustrated embodiment the ANC processing system **210** includes resources to form the digital noise reduction signal $F'(n)$ used to perform active noise cancellation. Alternatively, in some embodiments the digital noise reduction signal $F'(n)$ may be formed using resources within the audio processing system **220** of the audio device **104**.

FIG. 4 is a block diagram of an exemplary low latency ANC processing system **210** for performing active noise cancellation as described herein. In exemplary embodiments, the low latency ANC processing system **210** is embodied within a memory device within the ANC device **204**.

The low latency ANC processing system **210** may include analog-to-digital converter (A/D) module **400**, digital filter **410**, and digital-to-analog converter (D/A) module **420**. The low latency ANC processing system **210** may include more or fewer components than those illustrated in FIG. 4, and the functionality of modules may be combined or expanded into fewer or additional modules. Exemplary lines of communication are illustrated between various modules of FIG. 4, and in other figures herein. The lines of communication are not intended to limit which modules are communicatively coupled with others, nor are they intended to limit the number and type of signals communicated between modules.

In operation, the analog reference signal $r(t)$ generated by the reference microphone **106** is provided to oversampling data converter **406** within the A/D module **400**. The oversampling data converter **406** converts the analog reference signal

$r(t)$ into a digital reference signal $R(n)$ at a first sampling rate. In the illustrated embodiment, the digital reference signal $R(n)$ is a one-bit data stream at a sampling rate of 3.027 MHz or 2.288 MHz. Alternatively, other sampling rates and number of bits may be used.

As used herein, an “oversampling data converter” is an analog-to-digital converter with a sampling rate higher than the target sample rate, such as for example by a factor between 8 and 512. In other words, there exist multiple samples of signal $R(n)$ for each sample of signal $R'(n)$.

In the illustrated embodiment, the oversampling data converter **406** is a sigma-delta modulator. Alternatively, other types of oversampling data converters such as a flash converter may be used.

The digital reference signal $R(n)$ is provided to decimator module **408**. The decimator module **408** downsamples the digital reference signal $R(n)$ to produce a decimated digital reference signal $R'(n)$ at a second sampling rate less than the first sampling rate. In other words, the decimator module **408** downsamples the digital reference signal $R(n)$ by a predetermined downsampling factor (decimation factor) to form the decimated digital reference signal $R'(n)$. In the illustrated embodiment, the decimated digital reference signal $R'(n)$ has a sampling rate between 100 and 800 KHz. The decimator module **408** is described in more detail below with respect to FIG. 5.

The decimated digital reference signal $R'(n)$ is then filtered by digital filter **410** to form the digital noise reduction signal $F'(n)$. The digital filter **410** is based on a transfer function which models the acoustic path from the location of the reference microphone **106** to the first listening position **130**. The transfer function may incorporate characteristics of the acoustic path, such as one or more of an amplitude, phase shift and time delay from the reference microphone **106** to the first listening position **130**. The transfer function can also model the reference microphone **106** response, the audio transducer **116** response, and the acoustic path from the audio transducer **116** to the first listening position **130**.

The parameter values of the digital filter **410** may for example be determined empirically through calibration. The parameter values (e.g. filter gain and cutoff frequency) of the digital filter **410** may for example be adjusted from time to time. This adjustment may for example be in response to a feedback signal, as described in more detail below with reference to FIGS. 8 and 9. In such a case, the parameter values may for example be stored in the form of a look-up table stored in the memory within the ANC device **204**. As another example, the parameter values may be stored in the form of an approximate function derived based on the calibration measurements.

The decimated digital reference signal $R'(n)$ is also provided to optional decimator module **460**. The decimator module **460** further downsamples decimated digital reference signal $R'(n)$ to produce decimated digital reference signal $R''(n)$ at the target sampling rate. In the illustrated embodiment the decimator module **460** comprises a multi-stage half-band infinite impulse response (IIR) decimator. The decimation factor may be for example between 2, 4 and 8.

The D/A module **420** receives the digital noise reduction signal $F'(n)$. The D/A module **420** also receives the digital desired signal $S(n)$ from the audio device **104**. An interpolator module **422** within the D/A module **420** “interpolates” the digital desired signal $S(n)$ by upsampling its sampling rate to form interpolated digital desired signal $S'(n)$.

Combiner **426** then combines the digital noise reduction signal $F'(n)$ with the interpolated digital desired signal $S'(n)$ to form combined digital signal $G'(n)$. The combined digital

signal $G'(n)$ is then provided to the D/A converter **424**. The D/A converter **424** converts the digital output of the combiner **426** into an analog electric signal $g(t)$. The analog electric signal $g(t)$ is then provided to the audio transducer **116**. Active noise cancellation is then performed at the first listening position **130**, whereby the audio transducer **116** generates the acoustic wave **107** in response to the analog electric signal $g(t)$.

The latency introduced during decimation and interpolation of digital signals can be substantial. The present technology provides low latency ANC by decimating the digital reference signal $R(n)$ to a sampling rate for the decimated digital reference signal $R'(n)$ which is greater than the Nyquist sampling rate. As a result, the latency introduced by the decimation of the digital reference signal $R(n)$ can be significantly less than the latency introduced if the digital reference signal $R(n)$ were decimated down to the Nyquist sampling rate. In addition, by maintaining a relatively high sampling rate in the decimated digital reference signal $R'(n)$, the digital noise reduction signal $F'(n)$ “bypasses” the interpolation performed by the interpolator module **422**. As a result, the latency which would be introduced by the interpolator module **422** is avoided altogether. In doing so, in embodiments the latency of the ANC device **204** between receiving the primary acoustic wave **111** and generating the secondary acoustic wave **107** can be less than or equal to 100 microseconds. In some embodiments, this latency can be less than or equal to 50 microseconds.

FIG. 5 is a block diagram of an exemplary A/D module **400**. The oversampling data converter **406** may include a pre-gain amplifier (PGA) **500** and an analog sigma-delta modulator **502**. The decimator module **408** may include a cascaded integrated comb (CIC) decimator **504** and a multi-stage half-band infinite impulse response (IIR) decimator **506**. The oversampling data converter **406** and the decimator module **408** may each include more or fewer components than those illustrated in FIG. 5, and the functionality of modules may be combined or expanded into fewer or additional modules.

The pre-gain amplifier **500** applies a gain to the analog reference signal $r(t)$. The output of the pre-gain amplifier **500** is provided to the analog sigma-delta modulator **502**. The analog sigma-delta modulator **502** converts the weighted analog reference signal $r(t)$ into the digital reference signal $R(n)$ at a first sampling rate. The digital reference signal $R(n)$ is a sigma-delta modulator data stream which is typically a one-bit or very small number of bits data stream. As a result, it can be difficult to directly perform signal processing operations such as filtering directly on the digital reference signal $R(n)$. In particular, various signal processing techniques such as filtering first require conversion of the sigma-delta modulator data stream into a multi-bit pulse-code modulation (PCM) data stream. As described in more detail below, this conversion is performed by the decimator module **408**. Specifically, the decimator module **408** both downsamples the digital reference signal $R(n)$ and also generates a multi-bit PCM data stream on which subsequent signal processing steps can then be performed.

The CIC decimator **504** then downsamples the digital reference signal $R(n)$ by a first decimation factor. The first decimation factor may be for example between 1 and 32. The weighted multi-stage half-band infinite impulse response (IIR) decimator **506** then further decimates the output of the CIC decimator **504** by a second decimation factor to form the decimated digital reference signal $R'(n)$. The second decimation factor may be for example between 2 and 4. The CIC decimator **504** is advantageous because it provides a very

high sample rate, and it also very cheap. However, the frequency response of the CIC decimator **504** typically is not sufficient for forming a high quality final audio signal at the target sample rate. Including the IIR decimator **506** after the CIC decimator **504**, at a lower (cheaper in MIPS) rate, the combination can provide a higher quality overall frequency response for the final signal. In alternative embodiments the decimator module **408** may be different than that illustrated in FIG. 5. For example, other types of FIR (finite impulse response) decimation filters may be used at the highest rate by exploiting the fact that the incoming sigma-delta modulator data stream is typically only one bit. In such a case, a one bit multiplier can be implemented as an adder/subtractor combined with a simple table lookup for the polyphase coefficients. In preferred embodiments, an IIR halfband decimator may be used since it provides very low latency as well as a very low implementation cost in terms of memory and MIPS.

FIG. 6 is a block diagram of an exemplary D/A converter **424**. The D/A converter **424** may each include more or fewer components than those illustrated in FIG. 6, and the functionality of modules may be combined or expanded into fewer or additional modules.

The combined digital signal $G'(n)$ is provided to a multi-stage half-band IIR interpolator **600**. The multi-stage half-band IIR interpolator **600** interpolates the combined digital signal $G'(n)$ by upsampling its sampling rate by a first interpolation factor. The first interpolation factor may for example be 4. CIC interpolator **610** then further interpolates the output of the multi-stage half-band IIR interpolator **600** by a second interpolation factor. The second interpolation factor may be for example between 1 and 32.

Digital sigma-delta module **620** then quantizes (i.e. reduces the number of bits) the digital output of the CIC interpolator **610** and shapes the quantization noise. Digital-to-analog converter (DAC) **630** then converts the digital output of the CIC interpolator **610** into a corresponding analog signal. The analog signal is then filtered by analog reconstruction filter **640** to form the analog electric signal $g(t)$.

FIG. 7 is a flow chart of an exemplary method **700** for performing active noise cancellation. As with all flow charts herein, in some embodiments the steps may be combined, performed in parallel, or performed in a different order. The method **700** of FIG. 7 may also include addition or fewer steps than those illustrated.

In step **710**, the primary acoustic wave **111** is received by the reference microphone **106** to form analog reference signal $r(t)$. In some embodiments, more than one reference signal may be received and processed.

In step **720**, the analog reference signal $r(t)$ is converted into the digital reference signal $R(n)$ using the oversampling data converter **406**. In step **730**, decimation is performed on the digital reference signal $R(n)$ to form decimated digital reference signal $R'(n)$.

In step **740**, the digital noise reduction signal $F'(n)$ is formed by applying the digital filter **410** to the decimated digital reference signal $R'(n)$. In step **750**, the digital noise reduction signal $F'(n)$ is converted into an analog noise reduction signal to form analog electric signal $g(t)$. In step **760**, the analog electric signal $g(t)$ is then provided to the audio transducer **116** of the first earpiece **112** of the headset **120** to generate the secondary acoustic wave **107**, thereby performing active noise cancellation at the first listening position **130**.

FIG. 8 illustrates an expanded view of a second exemplary earpiece **112** which includes a monitoring microphone **806** which can be utilized to perform active noise cancellation as described herein. As shown in FIG. 8, the monitoring microphone **806** is located at a monitoring position within the

earpiece **112**. The signal received by the monitoring microphone **806** is referred to as monitoring signal $m(t)$. Due to the position of the monitoring microphone **806** within the earpiece **112**, the monitoring signal $m(t)$ indicates the acoustic energy level within the earpiece **112**.

The monitoring signal $m(t)$ can then be utilized by the ANC device **204** to adjust the parameters (e.g. filter gain and cutoff frequency) of the digital filter **410** used to form the digital noise reduction signal $F'(n)$. By adjusting the digital filter **410** based on the monitoring signal $m(t)$, the digital noise reduction signal $F'(n)$ can be adjusted so as to optimize noise cancellation at the first listening position **130**. By doing so, the active noise cancellation techniques described herein can achieve optimal noise cancellation in diverse and dynamic acoustic environments.

As used herein, a given signal, event or value is "based on" a predecessor signal, event or value if the predecessor signal, event or value influenced the given signal, event or value. If there is an intervening processing element, step or time period, the given signal can still be "based on" the predecessor signal, event or value. If the intervening processing element or step combines more than one signal, event or value, the output of the processing element or step is considered to be "based on" each of the signal, event or value inputs. If the given signal, event or value is the same as the predecessor signal, event or value, this is merely a degenerate case in which the given signal, event or value is still considered to be "based on" the predecessor signal, event or value. "Dependency" on a given signal, event or value upon another signal, event or value is defined similarly.

The above described modules may be comprised of instructions that are stored in a storage medium such as a machine readable medium (e.g., computer readable medium). These instructions may be retrieved and executed by a processor. Some examples of instructions include software, program code, and firmware. Some examples of storage media comprise memory devices and integrated circuits. The instructions are operational.

While the present invention is disclosed by reference to the preferred embodiments and examples detailed above, it is to be understood that these examples are intended in an illustrative rather than a limiting sense. It is contemplated that modifications and combinations will readily occur to those skilled in the art, which modifications and combinations will be within the spirit of the invention and the scope of the following claims.

What is claimed is:

1. A method for reducing an acoustic energy level at a listening position, the method comprising:
 - receiving a primary acoustic wave at a reference position to form an analog reference signal;
 - converting the analog reference signal into a digital reference signal at a first sample rate using an oversampling data converter, the oversampling data converter being for receiving the analog reference signal and converting the analog reference signal into the digital reference signal at the first sample rate;
 - forming a digital noise reduction signal based on the digital reference signal;
 - converting the digital noise reduction signal into an analog noise reduction signal; and
 - generating a secondary acoustic wave based on the analog noise reduction signal, the secondary acoustic wave adapted to reduce the acoustic energy level at the listening position.

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2. The method of claim 1, further comprising performing decimation to reduce the first sample rate of the digital reference signal to a second sample rate prior to forming the digital noise reduction signal.

3. The method of claim 2, wherein the digital noise reduction signal is at the second sample rate, and converting the digital noise reduction signal comprises performing interpolation to increase the second sample rate of the digital noise reduction signal to a third sample rate prior to converting the digital noise reduction signal into the analog noise reduction signal.

4. The method of claim 2, wherein the digital noise reduction signal is at the second sample rate, and further comprising:

receiving a digital desired signal at a third sample rate less than the second sample rate;
converting the digital desired signal into an analog desired signal; and
generating the secondary acoustic wave further based on the analog desired signal.

5. The method of claim 1, wherein the oversampling data converter is a sigma-delta modulator.

6. The method of claim 1, wherein a latency between receiving the primary acoustic wave and generating the secondary acoustic wave is less than or equal to 100 microseconds.

7. The method of claim 1, wherein the primary acoustic wave is received at the reference position by a reference microphone arranged on an earpiece of a headset, and the secondary acoustic wave is generated by an audio transducer arranged on the earpiece.

8. The method of claim 7, further comprising:

receiving the primary acoustic wave and the secondary acoustic wave via a monitoring microphone to form a monitoring signal, the monitoring microphone arranged between the audio transducer and the listening position; and
generating the secondary acoustic wave further based on the monitoring signal.

9. A system for reducing an acoustic energy level at a listening position, the system comprising:

a reference microphone to receive a primary acoustic wave at a listening position to form an analog reference signal; and
a noise cancellation module to:

convert the analog reference signal into a digital reference signal at a first sample rate using an oversampling data converter, the oversampling data converter being for receiving the analog reference signal and converting the analog reference signal into the digital reference signal;

form a digital noise reduction signal based on the digital reference signal; and

convert the digital noise reduction signal into an analog noise reduction signal; and

an audio transducer to generate a secondary acoustic wave based on the analog noise reduction signal, the secondary acoustic wave adapted to reduce the acoustic energy level at the listening position.

10. The system of claim 9, wherein the noise cancellation module further performs decimation to reduce the first sample rate of the digital reference signal to a second sample rate prior to forming the digital noise reduction signal.

11. The system of claim 10, wherein the digital noise reduction signal is at the second sample rate, and converting the digital noise reduction signal comprises performing interpolation to increase the second sample rate of the digital noise reduction signal to a third sample rate prior to converting the digital noise reduction signal into the analog noise reduction signal.

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12. The system of claim 10, wherein the digital noise reduction signal is at the second sample rate, and the noise cancellation module further receives a digital desired signal at a third sample rate less than the second sample rate and converts the digital desired signal into an analog desired signal, and the audio transducer to generate the secondary acoustic wave further based on the analog desired signal.

13. The system of claim 9, wherein the oversampling data converter is a sigma-delta modulator.

14. The system of claim 9, wherein a latency between receiving the primary acoustic wave and generating the secondary acoustic wave is less than or equal to 100 microseconds.

15. The system of claim 9, wherein the reference microphone and the audio transducer are each arranged on an earpiece of a headset.

16. The system of claim 15, further comprising a monitoring microphone to receive the primary acoustic wave and the secondary acoustic wave to form a monitoring signal, the monitoring microphone arranged between the audio transducer and the listening position, and wherein the noise cancellation module forms the digital noise reduction signal further based on the monitoring signal.

17. A non-transitory computer readable storage medium having embodied thereon a program, the program being executable using a processor to perform a method for reducing an acoustic energy level at a listening position, the method comprising:

receiving a primary acoustic wave at a reference position to form an analog reference signal;

converting the analog reference signal into a digital reference signal at a first sample rate using an oversampling data converter, the oversampling data converter being for receiving the analog reference signal and converting the analog reference signal into the digital reference signal;

forming a digital noise reduction signal based on the digital reference signal;

converting the digital noise reduction signal into an analog noise reduction signal; and

generating a secondary acoustic wave based on the analog noise reduction signal, the secondary acoustic wave adapted to reduce the acoustic energy level at the listening position.

18. The non-transitory computer readable storage medium of claim 17, wherein the method further comprises performing decimation to reduce the first sample rate of the digital reference signal to a second sample rate prior to forming the digital noise reduction signal.

19. The non-transitory computer readable storage medium of claim 17, wherein the digital noise reduction signal is at the second sample rate, and converting the digital noise reduction signal comprises performing interpolation to increase the second sample rate of the digital noise reduction signal to a third sample rate prior to converting the digital noise reduction signal into the analog noise reduction signal.

20. The non-transitory computer readable storage medium of claim 17, wherein the digital noise reduction signal is at the second sample rate, and the method further comprises:

receiving a digital desired signal at a third sample rate less than the second sample rate;

converting the digital desired signal into an analog desired signal; and

generating the secondary acoustic wave further based on the analog desired signal.